Perishable Lot–Sizing with Heterogeneous Shelf Lives

Compact MIP with Last–Expiry–First via No–Crossing Constraints

Indices and sets

- $T = \{1, 2, \ldots, \mathbf{T}\}$ — planning periods
- $I = \{1, 2, \ldots, \mathbf{I}\}$ — items
- $t \in T$ — production period, $u \in T$ — consumption period
- m_{it} — shelf life of item i produced at t (may vary by t)

For each item i define the feasible arc set

$$\mathcal{A}_i = \{\, (t, u) \in T \times T : t \leq u \leq t + m_{it} \,\}$$

and the expiry marker for production at t

$$v_{it} = t + m_{it}$$

Perishability

A lot produced at period t can be used only at periods u with $t \leq u \leq t + m_{it}$ and it expires at $v_{it} = t + m_{it}$

Parameters

d_{iu}	demand of item i at period u
c_{it}	unit production cost for item i at period t
h_{it}	holding cost per unit of age (constant or time varying)
s_{it}	setup cost for item i at period t
κ_t	total production capacity at period t
p_{it}	optional per–item capacity at period t
μ_{it}	setup linking constant $\mu_{it} = \sum_{u:(t,u)\in\mathcal{A}_i} d_{iu}$
g_{itu}	cost of producing in t and consuming in u defined as $c_{it} + h_{it}(u - t)$
C_{iu}	upper bound for flow into demand (i, u) (use $C_{iu} = d_{iu}$ for no backorders)

Decision variables

$X_{itu} \geq 0$	quantity of item i produced at t and consumed at u for $(t,u) \in \mathcal{A}_i$
$Y_{it} \in \{0,1\}$	setup decision for item i at t
$Z_{itu} \in \{0,1\}$	arc activation, equals 1 iff arc (t,u) is used

Objective

$$\min \sum_{i \in I} \sum_{(t,u) \in \mathcal{A}_i} g_{itu} \, X_{itu} + \sum_{i \in I} \sum_{t \in T} s_{it} \, Y_{it}$$

Constraints

Global capacity for all $t \in T$

$$\sum_{i \in I} \sum_{u:(t,u) \in \mathcal{A}_i} X_{itu} \leq \kappa_t$$

Optional per–item capacity for all $i \in I$ and $t \in T$

$$\sum_{u:(t,u) \in \mathcal{A}_i} X_{itu} \leq p_{it}$$

Setup linking for all $i \in I$ and $t \in T$

$$\sum_{u:(t,u) \in \mathcal{A}_i} X_{itu} \leq \mu_{it} \, Y_{it}$$

Demand satisfaction for all $i \in I$ and $u \in T$

$$\sum_{t:(t,u) \in \mathcal{A}_i} X_{itu} = d_{iu}$$

Arc activation link for all $i \in I$ and $(t,u) \in \mathcal{A}_i$

$$0 \leq X_{itu} \leq C_{iu} \, Z_{itu}$$

Last–expiry–first as no–crossing for all $i \in I$ and for all arc pairs $(t,s) \in \mathcal{A}_i$ and $(t',u) \in \mathcal{A}_i$ with $s < u$ and $v_{it} < v_{it'}$

$$Z_{its} + Z_{it'u} \leq 1$$

Variable domains

$$X_{itu} \geq 0, \quad Y_{it} \in \{0,1\}, \quad Z_{itu} \in \{0,1\}$$

Notes

- $v_{it} = t + m_{it}$ is the expiry period of production at t
- The link $X_{itu} \leq C_{iu} Z_{itu}$ turns on exactly those arcs that carry positive flow and keeps the

big–M tight when $C_{iu} = d_{iu}$

- The no–crossing rule enforces a last–expiry–first order across time: as consumption time increases, the sequence of chosen expiry markers $v_{i,t(\cdot)}$ is nonincreasing
- You can generate the no–crossing constraints only for truly crossable pairs to keep the count modest